Changes made

- ros2 control config
    - Wheel radius measurement corrected to 0.175m. This measurement was physically verified by independent measurements from different team members.
    - Track width adjusted to new precise measurement.
- URDF
    - Base width re-measured and corrected.
    - Wheel_yoff re-measured and corrected. *The URDF and control config disagree by 2.3mm about the track width, which can be fixed by calibration, but I don't expect this miniscule error to matter too much.*
    - Wheel radius corrected to match ros2 control config.
- Rebuilt the workspace

Straight-line test

25ft was measured out and the robot was placed with the wheel center at x=0ft. It was teleop-driven forward in a straight line until the wheel center was at x=25ft. The navigation / perception system was running even though no autonomous goals were given.

Two runs of this test were performed. Bag files are real_bag1 and real_bag2. The robot was not kept perfectly straight the entire time and there was likely some precision error, but we tried to ensure the robot ended up in the right spots.

Autonomous turning test

A similar, easier course was assembled and driven autonomously. The main purpose of this was to compare the gyro yaw with the odometry yaw.

Run #1 of this is bag 3, and run #2 is bag 4.

Two runs of this test were performed. On run 1, the robot was teleop-driven back to its starting position after it reached the goal at a high speed. I assume this will also be useful.

In run 2, we attempted a goal inside our lab, through the very narrow passageway, and it failed. You don't have to comment too much on this; this is much harder than what we expect in the competition, but it might still be useful to understand why it failed even though the path *was* wide enough.

Bag command

ros2 bag record -o real_bag4 /diffbot_base_controller/cmd_vel /cmd_vel_raw /diffbot_base_controller/odom /odometry/filtered/local /odometry/filtered/global /global_costmap/costmap /local_costmap/costmap /velodyne_points_nav /imu/data /plan /odom_goal_pose /odom_goal_pose_rviz /tf /dynamic_joint_states /tf_static /local_costmap/published_footprint /global_costmap/published_footprint /rosout /behavior_tree_log /speed_limit /speed_governor/info